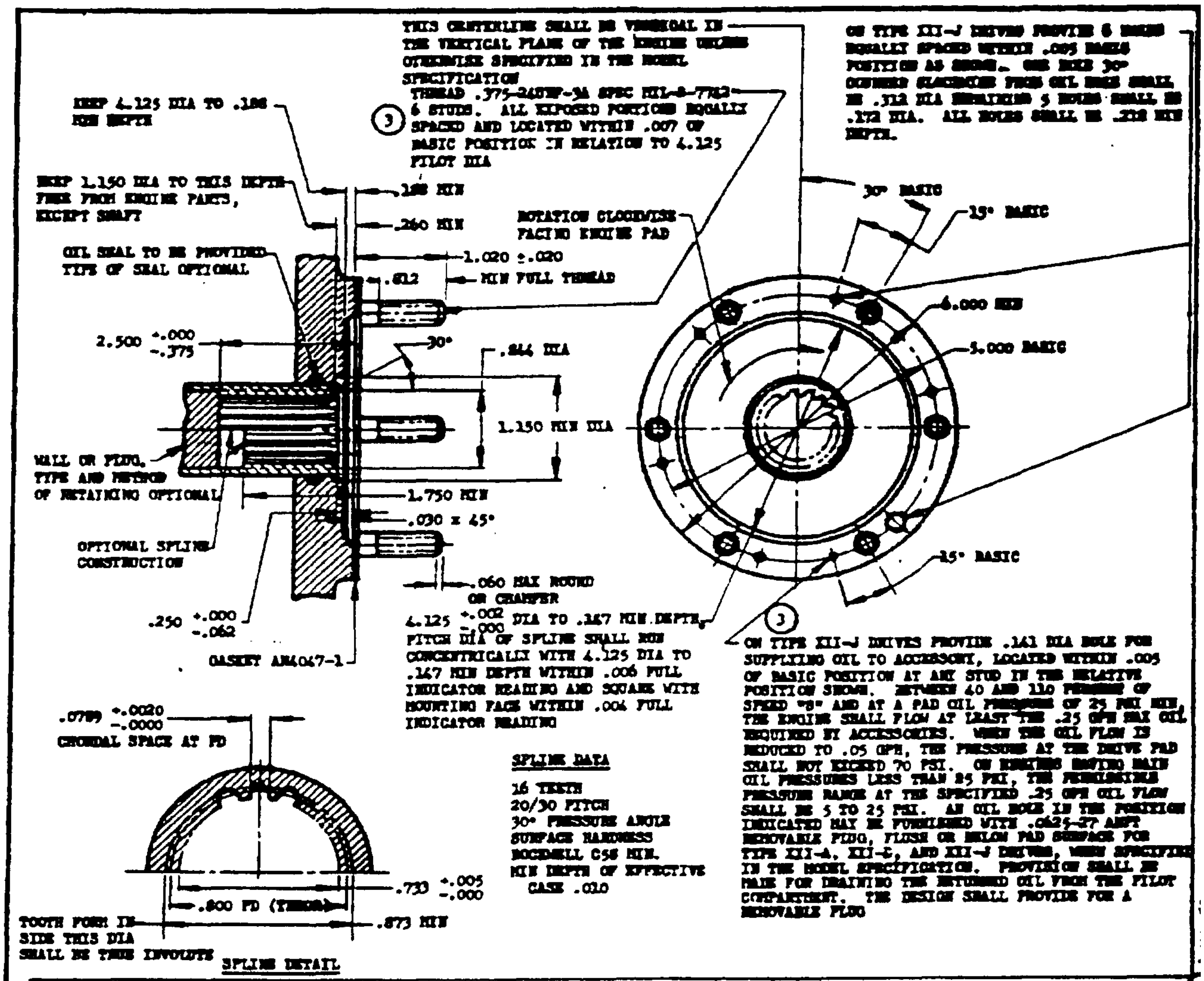
KEEP 4.125 DIA TO .188 MIN DEPTH
KEEP 1.150 DIA TO THIS DEPTH FREE FROM ENGINE PARTS, EXCEPT SHAFT
.188 MIN
.260 MIN
OIL SEAL TO BE PROVIDED TYPE OF SEAL OPTIONAL
1.020 ±.020
.812
MIN FULL THREAD
2.500 +.000 −.375
30°
.844 DIA
1.150 MIN DIA
WALL OR PLUG, TYPE AND METHOD OF RETAINING OPTIONAL
1.750 MIN
.030 x 45°
OPTIONAL SPLINE CONSTRUCTION
.060 MAX ROUND OR CHAMFER
.250 +.000 −.062
GASKET AN4047-1
ROTATION CLOCKWISE FACING ENGINE PAD
30° BASIC
15° BASIC
6.000 MIN
5.000 BASIC
15° BASIC
③
.0789 +.0020 −.0000 CHORDAL SPACE AT PD
.733 +.005 −.000
.800 PD (THEOR)
.873 MIN
TOOTH FORM INSIDE THIS DIA SHALL BE TRUE INVOLUTE
SPLINE DETAIL

THIS CENTERLINE SHALL BE VERTICAL IN THE VERTICAL PLANE OF THE ENGINE UNLESS OTHERWISE SPECIFIED IN THE MODEL SPECIFICATION

③ THREAD .375-24UNF-3A SPEC MIL-S-7742 6 STUDS. ALL EXPOSED PORTIONS EQUALLY SPACED AND LOCATED WITHIN .007 OF BASIC POSITION IN RELATION TO 4.125 PILOT DIA

ON TYPE XII-J DRIVES PROVIDE 6 HOLES EQUALLY SPACED WITHIN .005 BASIC POSITION AS SHOWN. ONE HOLE 30° COUNTER CLOCKWISE FROM OIL HOLE SHALL BE .312 DIA REMAINING 5 HOLES SHALL BE .172 DIA. ALL HOLES SHALL BE .218 MIN DEPTH.

4.125 +.002 −.000 DIA TO .147 MIN DEPTH. PITCH DIA OF SPLINE SHALL RUN CONCENTRICALLY WITH 4.125 DIA TO .147 MIN DEPTH WITHIN .006 FULL INDICATOR READING AND SQUARE WITH MOUNTING FACE WITHIN .004 FULL INDICATOR READING

③ ON TYPE XII-J DRIVES PROVIDE .141 DIA HOLE FOR SUPPLYING OIL TO ACCESSORY, LOCATED WITHIN .005 OF BASIC POSITION AT ANY STUD IN THE RELATIVE POSITION SHOWN. BETWEEN 40 AND 110 PERCENT OF SPEED "S" AND AT A PAD OIL PRESSURE OF 25 PSI MIN, THE ENGINE SHALL FLOW AT LEAST THE .25 GPH MAX OIL REQUIRED BY ACCESSORIES. WHEN THE OIL FLOW IS REDUCED TO .05 GPH, THE PRESSURE AT THE DRIVE PAD SHALL NOT EXCEED 70 PSI. ON ENGINES HAVING MAIN OIL PRESSURES LESS THAN 25 PSI, THE PERMISSIBLE PRESSURE RANGE AT THE SPECIFIED .25 GPH OIL FLOW SHALL BE 5 TO 25 PSI. AN OIL HOLE IN THE POSITION INDICATED MAY BE FURNISHED WITH .0625-27 ANPT REMOVABLE PLUG, FLUSH OR BELOW PAD SURFACE FOR TYPE XII-A, XII-C, AND XII-J DRIVES, WHEN SPECIFIED IN THE MODEL SPECIFICATION. PROVISION SHALL BE MADE FOR DRAINING THE RETURNED OIL FROM THE PILOT COMPARTMENT. THE DESIGN SHALL PROVIDE FOR A REMOVABLE PLUG

SPLINE DATA

16 TEETH
20/30 PITCH
30° PRESSURE ANGLE
SURFACE HARDNESS
ROCKWELL C58 MIN.
MIN DEPTH OF EFFECTIVE
CASE .010

TYPE	SPEED, STRENGTH, ACCESSORY DATA, AND USE								
	TORQUE LB -IN.			"S" SPEED RPM	ACCESSORY WEIGHT LB MAX	OVERHUNG MOMENT ③ LB-IN. MAX	③ ACCESSORY CLEARANCE AS APPLICABLE	NORMAL USE	TYPE OF ENGINE
	T_c	T_o	T_s						
XII-A	500	750	2200	7500-8250	65	400	AND10305	GENERATOR	RECIPROCATING
XII-D	500	750	2200	(a) 6000	65	500	AND10305	GENERATOR	TURBOJET. AND TURBOPROP
XII-J	600	900	2700	3250-3750	40	350	(b) ③	LOW SPEED FLUID POWER PUMPS ③	ALL ENGINES

(a) +400 −000 RPM WHEN THE ENGINE IS OPERATED AT 75% NORMAL SEA LEVEL STATIC OUTPUT.

③ (b) ACCESSORY CLEARANCE SHALL BE PROVIDED IN ACCORDANCE WITH MANUFACTURER'S MODEL SPECIFICATION.

③ FLANGE ADAPTER AND SPLINE ADAPTER IN ACCORDANCE WITH AN4138-1A AND AN4141-2, RESPECTIVELY, SHALL BE FURNISHED ON TYPE XII-J DRIVES FOR CONVERTING TO A TYPE XI DRIVE WHEN SPECIFIED IN THE ENGINE MODEL SPECIFICATION.

PAD OUTLINE AS DIMENSIONED IS A BASIC MINIMUM AREA REQUIREMENT.

THE OIL LEAKAGE OUT OF THIS DRIVE SHALL NOT EXCEED 2CC PER HOUR.

③ SPEED: "S" DRIVE SHAFT SPEED AT NORMAL RATED ENGINE SPEED FOR RECIPROCATING ENGINES AND AT 75% NORMAL SEA LEVEL STATIC OUTPUT FOR TURBINE ENGINES.

STRENGTH: TO BE CAPABLE OF DRIVING CONTINUOUS TORQUE LOAD (T_c) AT ANY ENGINE SPEED. THE DRIVE SHALL WITHSTAND STATIC TORQUE (T_s) WITHOUT FAILURE OR PERMANENT DEFORMATION. IN ADDITION, THE DRIVES SHALL BE ADEQUATE FOR FIVE (5) MINUTE PERIODS OF TORQUE OVERLOAD (T_o) OPERATION, WHICH OVERLOAD PERIODS SHALL BE CONSIDERED AS RECURRING AT FOUR (4) HOUR INTERVALS.

REMOVE ALL BURRS AND SHARP EDGES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES [illegible].

③ THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS.

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

DRIVE - TYPE XII ENGINE ACCESSORY

AND20002

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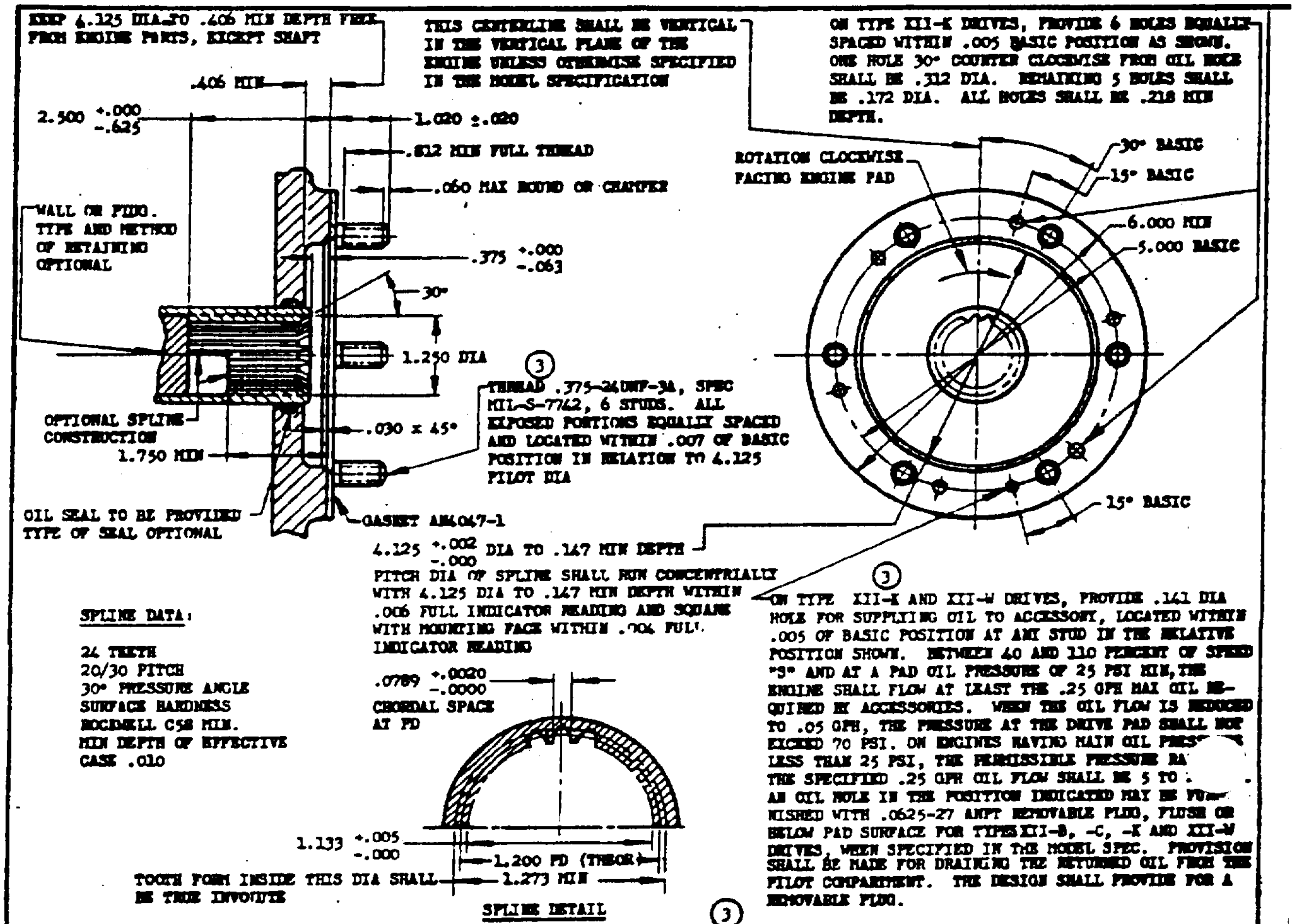
KEEP 4.125 DIA TO .406 MIN DEPTH FREE FROM ENGINE PARTS, EXCEPT SHAFT
.406 MIN
2.500 +.000 −.625
1.020 ±.020
.812 MIN FULL THREAD
.060 MAX ROUND OR CHAMFER
WALL OR PLUG. TYPE AND METHOD OF RETAINING OPTIONAL
.375 +.000 −.063
30°
1.250 DIA
③
THREAD .375-24UNF-3A, SPEC MIL-S-7742, 6 STUDS. ALL EXPOSED PORTIONS EQUALLY SPACED AND LOCATED WITHIN .007 OF BASIC POSITION IN RELATION TO 4.125 PILOT DIA
OPTIONAL SPLINE CONSTRUCTION
1.750 MIN
.030 x 45°
OIL SEAL TO BE PROVIDED TYPE OF SEAL OPTIONAL
GASKET AN4047-1
4.125 +.002 −.000 DIA TO .147 MIN DEPTH
THIS CENTERLINE SHALL BE VERTICAL IN THE VERTICAL PLANE OF THE ENGINE UNLESS OTHERWISE SPECIFIED IN THE MODEL SPECIFICATION
ON TYPE XII-K DRIVES, PROVIDE 6 HOLES EQUALLY SPACED WITHIN .005 BASIC POSITION AS SHOWN. ONE HOLE 30° COUNTER CLOCKWISE FROM OIL HOLE SHALL BE .312 DIA. REMAINING 5 HOLES SHALL BE .172 DIA. ALL HOLES SHALL BE .218 MIN DEPTH.
ROTATION CLOCKWISE FACING ENGINE PAD
30° BASIC
15° BASIC
6.000 MIN
5.000 BASIC
15° BASIC
PITCH DIA OF SPLINE SHALL RUN CONCENTRIALLY WITH 4.125 DIA TO .147 MIN DEPTH WITHIN .006 FULL INDICATOR READING AND SQUARE WITH MOUNTING FACE WITHIN .004 FULL INDICATOR READING
SPLINE DATA:
24 TEETH
20/30 PITCH
30° PRESSURE ANGLE
SURFACE HARDNESS ROCKWELL C58 MIN.
MIN DEPTH OF EFFECTIVE CASE .010
.0789 +.0020 −.0000
CHORDAL SPACE AT PD
1.133 +.005 −.000
1.200 PD (THEOR)
1.273 MIN
TOOTH FORM INSIDE THIS DIA SHALL BE TRUE INVOLUTE
SPLINE DETAIL

③ ON TYPE XII-K AND XII-W DRIVES, PROVIDE .141 DIA HOLE FOR SUPPLYING OIL TO ACCESSORY, LOCATED WITHIN .005 OF BASIC POSITION AT ANY STUD IN THE RELATIVE POSITION SHOWN. BETWEEN 40 AND 110 PERCENT OF SPEED "S" AND AT A PAD OIL PRESSURE OF 25 PSI MIN, THE ENGINE SHALL FLOW AT LEAST THE .25 GPH MAX OIL REQUIRED BY ACCESSORIES. WHEN THE OIL FLOW IS REDUCED TO .05 GPH, THE PRESSURE AT THE DRIVE PAD SHALL NOT EXCEED 70 PSI. ON ENGINES HAVING MAIN OIL PRES[illegible] LESS THAN 25 PSI, THE PERMISSIBLE PRESSURE RA[illegible] THE SPECIFIED .25 GPH OIL FLOW SHALL BE 5 TO [illegible] AN OIL HOLE IN THE POSITION INDICATED MAY BE FURNISHED WITH .0625-27 ANPT REMOVABLE PLUG, FLUSH OR BELOW PAD SURFACE FOR TYPES XII-B, -C, -K AND XII-W DRIVES, WHEN SPECIFIED IN THE MODEL SPEC. PROVISION SHALL BE MADE FOR DRAINING THE RETURNED OIL FROM THE PILOT COMPARTMENT. THE DESIGN SHALL PROVIDE FOR A REMOVABLE PLUG.

③

SPEED, STRENGTH, ACCESSORY DATA AND USE									
TYPE	TORQUE LB-IN.			"S" SPEED - RPM	ACCESSORY WEIGHT LB MAX	OVERHUNG MOMENT LB-IN. MAX	ACCESSORY CLEARANCE AS APPLICABLE	NOMINAL USE	TYPE OF ENGINE
	T_c	T_o	T_s						
XII-B	1000	1500	4 400	7500 - 8250	65	400	AND10305	GENERATOR	RECIPROCATING
XII-C	1500	2250	6 600	7500 - 8250	65	400	AND10305 (b)	GENERATOR POWER TAKEOFF	RECIPROCATING
XII-E	1000	1500	4 400	(a) 6000	75	625	AND10305	GENERATOR	TURBOJET AND TURBOPROP
XII-F	1500	2250	6 600	(a) 6000	75	625	AND10305 (b)	GENERATOR POWER TAKEOFF	TURBOJET AND TURBOPROP
XII-K	1000	1500	4 400	3250 - 3750	50	400	(b)	LOW SPEED FLUID POWER PUMPS ③	ALL
XII-M	2500	3750	11 000	7500 - 8250	65	400	AND10305 (b)	GENERATOR POWER TAKEOFF	RECIPROCATING
XII-N	2500	3750	11 000	(a) 6000	75	625	AND10305 (b)	GENERATOR POWER TAKEOFF	TURBOJET AND TURBOPROP
③ XII-W	1500	2250	6 600	3250 - 3750	75	625	(b)	LOW SPEED FLUID POWER PUMPS	ALL

(a) +400 - 000 RPM WHEN THE ENGINE IS OPERATING AT 75% NORMAL SEA LEVEL STATIC OUTPUT.

③ (b) ACCESSORY CLEARANCE SHALL BE PROVIDED IN ACCORDANCE WITH MANUFACTURER'S MODEL SPECIFICATION. SPLINE ADAPTER IN ACCORDANCE WITH AN4052-1 SHALL BE FURNISHED WHEN SPECIFIED IN THE ENGINE MODEL SPECIFICATION. FLANGE ADAPTER AND SPLINE ADAPTER IN ACCORDANCE WITH AN4138-1A AND AN4141-3, RESPECTIVELY, SHALL BE FURNISHED ON TYPE XII-K AND XII-W DRIVES FOR CONVERTING TO A TYPE XI DRIVE WHEN SPECIFIED IN THE ENGINE MODEL SPECIFICATION.

PAD OUTLINE AS DIMENSIONED IS A BASIC MINIMUM AREA REQUIREMENT.

THE OIL LEAKAGE OUT OF THIS DRIVE SHALL NOT EXCEED 2 CC PER HOUR.

③ SPEED: "S" DRIVE SHAFT SPEED AT NORMAL RATED ENGINE SPEED FOR RECIPROCATING ENGINES AND AT 75% NORMAL SEA LEVEL STATIC OUTPUT FOR TURBINE ENGINES.

STRENGTH: TO BE CAPABLE OF DRIVING CONTINUOUS TORQUE LOAD (T_c) AT ANY ENGINE SPEED. THE DRIVE SHALL WITHSTAND STATIC TORQUE (T_s) WITHOUT FAILURE OR PERMANENT DEFORMATION. IN ADDITION, THE DRIVES SHALL BE ADEQUATE FOR FIVE (5) MINUTE PERIODS OF TORQUE OVERLOAD (T_o) OPERATION, WHICH OVERLOAD PERIODS SHALL BE CONSIDERED AS RECURRING AT FOUR (4) HOUR INTERVALS.

REMOVE ALL BURRS AND SHARP EDGES.

③ DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±2°.

③ THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATION, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS.

APPROVED 7 May 47 REVISED ① 15 Jul 49 ② 14 Feb 52 ③ 4 Mar 53

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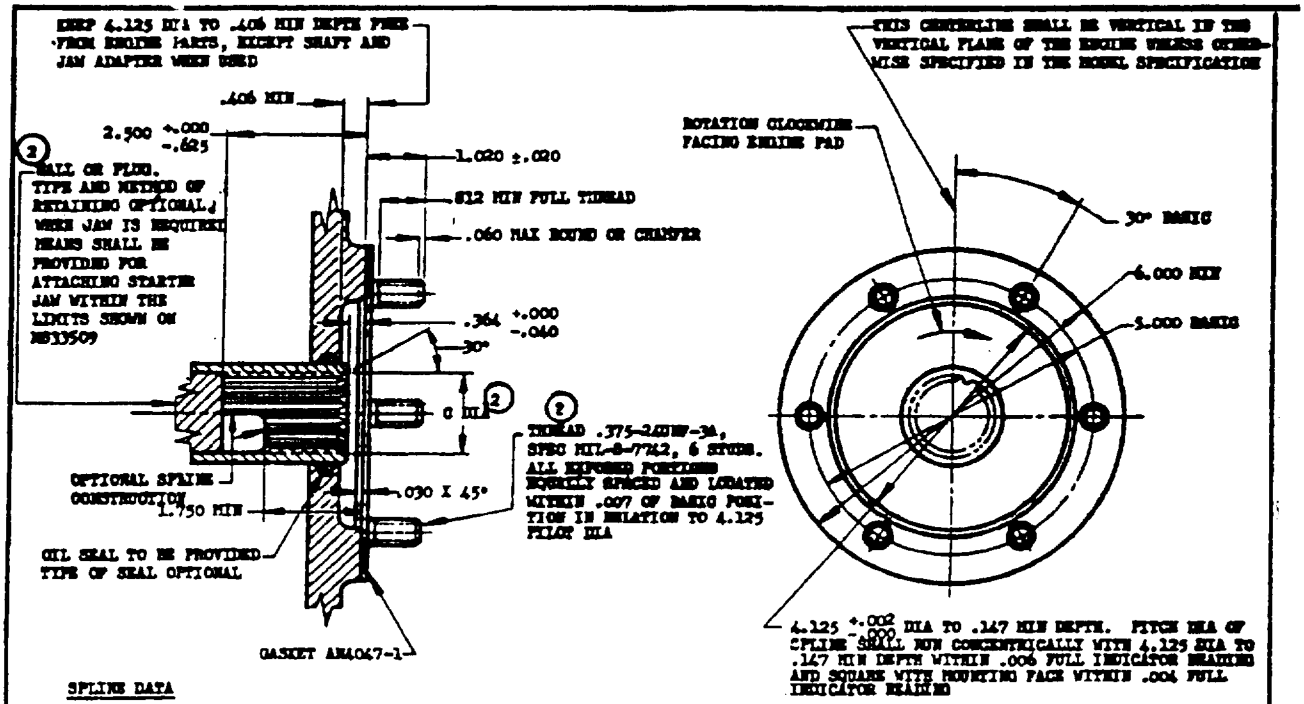
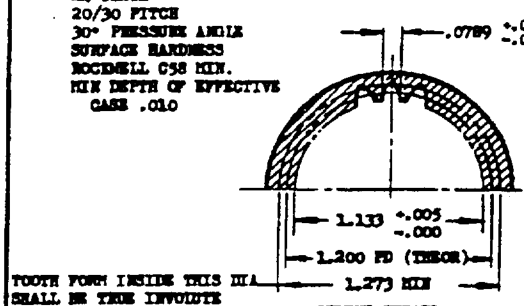
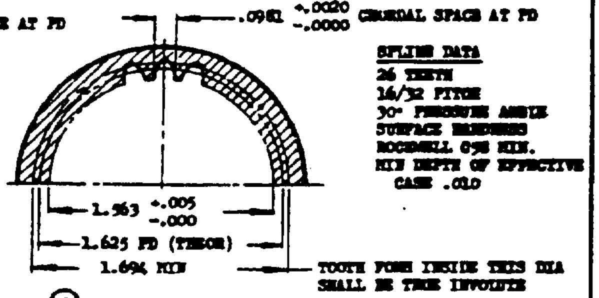
NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.


KEEP 4.125 DIA TO .406 MIN DEPTH FREE FROM ENGINE PARTS, EXCEPT SHAFT AND JAW ADAPTER WHEN USED
.406 MIN
2.500 +.000 -.625
② BALL OR PLUG. TYPE AND METHOD OF RETAINING OPTIONAL, WHEN JAW IS REQUIRED MEANS SHALL BE PROVIDED FOR ATTACHING STARTER JAW WITHIN THE LIMITS SHOWN ON MS33509
1.020 ±.020
.812 MIN FULL THREAD
.060 MAX ROUND OR CHAMFER
.364 +.000 -.040
30°
C DIA ②
② THREAD .375-24UNF-3A, SPEC MIL-S-7742, 6 STUDS. ALL EXPOSED PORTIONS EQUALLY SPACED AND LOCATED WITHIN .007 OF BASIC POSITION IN RELATION TO 4.125 PILOT DIA
OPTIONAL SPLINE CONSTRUCTION
1.750 MIN
.030 X 45°
OIL SEAL TO BE PROVIDED TYPE OF SEAL OPTIONAL
GASKET AN4047-1
THIS CENTERLINE SHALL BE VERTICAL IN THE VERTICAL PLANE OF THE ENGINE UNLESS OTHERWISE SPECIFIED IN THE MODEL SPECIFICATION
ROTATION CLOCKWISE FACING ENGINE PAD
30° BASIC
6.000 MIN
5.000 BASIC
4.125 +.002 -.000 DIA TO .147 MIN DEPTH. PITCH DIA OF SPLINE SHALL RUN CONCENTRICALLY WITH 4.125 DIA TO .147 MIN DEPTH WITHIN .006 FULL INDICATOR READING AND SQUARE WITH MOUNTING FACE WITHIN .004 FULL INDICATOR READING



SPLINE DETAIL



② SPLINE DETAIL

SPEED, STRENGTH, ACCESSORY DATA, AND USE								
TYPE	② SPLINE PD	STRENGTH	(a) "S" SPEED RPM	ACCESSORY WEIGHT LB MAX	OVERHUNG ② MOMENT LB-IN. MAX	NOMINAL USE	TYPE OF ENGINE	② C CHAMFER DIA
XII-S	1.200	(b)	3300 ±300	75	625	STARTER	TURBOJET AND TURBOPROP	1.250
XII-T	1.625	(c)	3300 ±300	75	625	STARTER		1.656

② (a) SPEED: "S" SPEED IS THE DRIVE SHAFT SPEED ABOVE WHICH THE ENGINE SHALL ACCELERATE TO IDLE SPEED WITHIN FIVE (5) SECONDS AT STATIC SEA LEVEL STANDARD CONDITIONS WITHOUT ASSISTANCE FROM THE STARTER.

② (b) DESIGN STRENGTH: FOR TURBOJET ENGINES THE DRIVE SHALL BE CAPABLE OF WITHSTANDING THE MAXIMUM TORQUE EQUAL TO 2.0 TIMES THE CONSTANT TORQUE REQUIRED FOR A FIFTEEN (15) SECOND STARTING PERIOD BEGINNING AT REST AND ENDING AT DRIVE SPEED "S".

FOR TURBOPROP ENGINES THE DRIVE SHALL BE CAPABLE OF WITHSTANDING THE MAXIUM TORQUE EQUAL TO 2.0 TIMES THE CONSTANT TORQUE REQUIRED FOR A TWENTY FIVE (25) SECOND STARTING PERIOD BEGINNING AT REST AND ENDING AT DRIVE SPEED "S".

(c) THE DRIVE SHALL BE CAPABLE OF WITHSTANDING EXTREMELY FAST STARTS AND THE STRENGTH SHALL BE AS SPECIFIED IN THE ENGINE MODEL SPECIFICATION.

JAW ADAPTER SHALL BE FURNISHED FOR CONVERTING TO A JAW TYPE DRIVE WHEN SO SPECIFIED IN THE ENGINE MODEL SPECIFICATION. WHEN FURNISHED, THE JAW CONFIGURATION AND INSTALLATION SHALL BE AS SHOWN ON DRAWING MS33509.

PAD OUTLINE AS DIMENSIONED IS A BASIC MINIMUM AREA REQUIREMENT.

THE OIL LEAKAGE OUT OF THIS DRIVE SHALL NOT EXCEED 200 PER HOUR. OVERBOARD DRAINS PERMITTED.

② ACCESSORY CLEARANCE SHALL BE PROVIDED IN ACCORDANCE WITH MANUFACTURER'S MODEL SPECIFICATION.

REMOVE ALL BURRS AND SHARP EDGES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±2°.

② THIS DRAWING AND THE APPLICABLE ENGINE SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS.

APPROVED 14 Feb 51 REVISED ① 17 Jan 52 ② 4 Mar 53

NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all associated standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than 6 months after the latest date of approval shown.

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DRIVE - TYPE XII ENGINE ACCESSORY

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NOT A PART NUMBER